

FastAPI Q1 Answers

1. What is CRUD in FastAPI?

CRUD stands for:

- **Create** → Add new data (POST)
- **Read** → Get data (GET)
- **Update** → Modify data (PUT/PATCH)
- **Delete** → Remove data (DELETE)

In FastAPI, CRUD operations are used to manage data using APIs.

2. Why do we use POST method for inserting data?

POST is used to send data to the server and create a new record.

Because:

- Data is sent in request body
- It is not visible in URL
- It is secure for creating resources

3. Difference between PUT and PATCH

PUT	PATCH
Replaces full record	Updates partial data
Must send all fields	Only required fields
Example: update entire student	Example: update only marks

4. What is Path Parameter?

Path parameter is a value passed in URL.

Example:

```
GET /students/1
```

Here 1 is a path parameter used to identify a student.

5. What is Request Body in FastAPI?

Request body is data sent in JSON format inside POST/PUT requests.

Example:

```
{  
  "name": "Rahul",  
  "age": 21  
}
```

It is used to send structured input data.

6. Why do we use Pydantic models?

Pydantic is used for:

- Data validation
- Type checking
- Automatic error handling
- Schema generation in Swagger

Example:

```
class Student(BaseModel):  
    name: str  
    age: int
```

7.What status code is returned for successful deletion?

For successful deletion:

200 OK

Sometimes also:

204 No Content

8.How can we check whether a student exists before updating?

We check using a loop:

```
for student in students:  
    if student.id == id:  
        # student exists
```

If not found → return:

```
HTTPException(status_code=404)
```

9.What happens if duplicate IDs are inserted?

If duplicate IDs are inserted:

- Data inconsistency occurs
- Multiple records with same ID exist
- Updates/deletes may affect wrong record

So we should validate ID before inserting.

10. How does FastAPI automatically generate Swagger UI?

FastAPI automatically generates Swagger UI using:

- Function signatures
- Pydantic models
- Type hints

It creates documentation at:

<http://127.0.0.1:8000/docs>

This is powered by OpenAPI standard.